

ORION

PART 10

Data & Intelligence Fabric

Telemetry, event streams, databases, analytics and information flow

Technical Architecture Series • Beginner → Professional

How to use this volume

This document is one independent volume in the twelve-part ORION technical series. Beginners should read from the start; experienced readers can jump directly to the relevant architecture sections.

ORION is a secure communications, infrastructure-coordination, emergency-response, and decision-support architecture. High-impact weapons-control functionality is deliberately outside this design.

Purpose

Move telemetry, events, identity records, incidents, operational state, analytics, and audit information through reliable, governed pipelines.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Data flow

Devices/services → telemetry collectors → event bus → stream processing + operational databases + object storage → analytics/AI/twin → dashboards and decisions.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Data categories

Operational data, telemetry, events, audit records, configuration, identity, incidents, analytics datasets, and simulation datasets.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Storage choices

Relational databases for structured transactions; NoSQL for suitable high-volume/flexible workloads; object storage for large files; warehouses for analytics.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Consistency

Define consistency per data class. Authorization state may need stronger guarantees than aggregated telemetry.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Event streaming

Use schemas, versioning, timestamps, unique IDs, correlation IDs, and duplicate/reordering tolerance where appropriate.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Governance

Ownership, classification, retention, access control, quality, lineage, schema evolution, deletion, and privacy rules.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

AI data

Separate training/evaluation data from production operational data. Minimize sensitive information and govern access explicitly.

Professional lens: define ownership, interfaces, trust boundaries, failure behavior, security controls, observability, and measurable success criteria before implementation.

Architecture Review Checklist

- Requirements and assumptions are explicit.
- Component ownership and boundaries are documented.
- Normal, degraded, and failure states are defined.
- Security and privacy are architectural requirements.
- Capacity and scaling assumptions are measurable.
- Logs, metrics, traces, and audit events cover critical flows.
- Dependencies have explicit contracts and failure behavior.
- Important decisions record alternatives and trade-offs.
- Testing covers expected behavior and controlled failures.
- Operational runbooks exist for important incidents.

Key Takeaways

Data & Intelligence Fabric should be implemented as a bounded layer of ORION, with explicit interfaces to the rest of the platform.

The goal is not maximum complexity. The goal is measurable resilience, security, maintainability, interoperability, and operational usefulness.

Glossary

- API — Application Programming Interface used for software-to-software communication.
- Edge — Computing placed near users or physical systems.
- Event — A record that something happened.
- Failover — Switching to a redundant component after failure.
- Idempotency — Repeating an operation does not create an unintended additional effect.
- NTN — Non-terrestrial network, commonly involving satellite infrastructure.
- Policy — A rule that determines what is allowed under stated conditions.
- RPO — Recovery Point Objective; acceptable data-loss window.
- RTO — Recovery Time Objective; target recovery time.
- Zero trust — Continuous verification instead of implicit trust based on network location.